

Notes of Modern Reinforcement Learning / Policy Learning Methods

Yue Wu

September 13, 2026

Outline

Intro

Smart-conservative-greedy parameterized algorithm

Intro

(2002) Conservative policy iteration

(2015) Trust Region Policy Optimization (TRPO)

(2017) Proximal Policy Optimization (PPO)

Policy learning for LLM

Intro

(2022) Reinforcement Learning with Human Feedback (RLHF)

(2023) Direct Policy Optimization (DPO)

(2024) Group Relative Policy Optimization (GRPO)

Beyond MDP

(2021) Decision Transformer

Discussion

On/Off-line and On/Off-policy

Outline

Intro

Smart-conservative-greedy parameterized algorithm

Intro

(2002) Conservative policy iteration

(2015) Trust Region Policy Optimization (TRPO)

(2017) Proximal Policy Optimization (PPO)

Policy learning for LLM

Intro

(2022) Reinforcement Learning with Human Feedback (RLHF)

(2023) Direct Policy Optimization (DPO)

(2024) Group Relative Policy Optimization (GRPO)

Beyond MDP

(2021) Decision Transformer

Discussion

On/Off-line and On/Off-policy

Intro

This note mainly focus on some modern RL methods, including

- ▶ Q-value based method
 - ▶ Deep Q Network
 - ▶ Conservative Q-learning
- ▶ Policy based method
 - ▶ TRPO
 - ▶ PPO
 - ▶ ...
- ▶ Decision Transformer (Without MDP assumption)

Outline

Intro

Smart-conservative-greedy parameterized algorithm

Intro

(2002) Conservative policy iteration

(2015) Trust Region Policy Optimization (TRPO)

(2017) Proximal Policy Optimization (PPO)

Policy learning for LLM

Intro

(2022) Reinforcement Learning with Human Feedback (RLHF)

(2023) Direct Policy Optimization (DPO)

(2024) Group Relative Policy Optimization (GRPO)

Beyond MDP

(2021) Decision Transformer

Discussion

On/Off-line and On/Off-policy

Intro: Smart-conservative-greedy parameterized algorithm

- ▶ Conservative greedy policy improvement algorithm
 - ▶ every iteration adds a new policy to maintain, costing a lot of space
 - ▶ the greedy policy part may be hard to solve
- ▶ Trust Region Policy Optimization (TRPO)
 - ▶ uses a parametric form of policy, just need to maintain and update the policy parameter
 - ▶ can be seemed as a smart-conservative-greedy algorithm by **bounding the difference between old and new policy**
 - ▶ theoretically good
- ▶ Proximal Policy Optimization (PPO)
 - ▶ also uses a parametric form of policy
 - ▶ similar object to maximize as TRPO, but use its pessimistic bound, **clip on the goal directly instead of using KL-divergence constraint**
 - ▶ compared to TRPO, a little bit ad hoc, but more technically efficient, easy to implement

Performance difference lemma and Policy Improvement Theorem

(Kakade and Langford 2002)

[Performance difference lemma]



[Policy Improvement Theorem]



Trust Region Policy Optimization (TRPO)

(Schulman, Levine, et al. 2015)

TRPO extends Policy Improvement Theorem by replacing α with a distance measure between π and $\tilde{\pi}$.

$$\alpha = D_{TV}^{\max}(\pi, \pi^{\text{new}})$$

This theorem quantifies the least change of value for any two policies $\pi, \tilde{\pi}$ with policy advantage and KL-divergence of the two policies,

$$\eta(\tilde{\pi}) - \eta(\pi) \geq A_{\pi}(\tilde{\pi}) - \frac{2\gamma\varepsilon}{(1-\gamma)^2} D_{KL}^{\max}(\tilde{\pi}, \pi)$$

So we can design an **Approximate** Policy Iteration algorithm that guarantees increasing value:

In each iteration $k = 0, 1, 2, \dots$ Solve the constrained optimization problem

$$\pi_{k+1} = \arg \max_{\pi} \left[A_{\pi_k}(\pi) - \frac{2\gamma\varepsilon}{(1-\gamma)^2} D_{KL}^{\max}(\pi_k, \pi) \right]$$

Trust Region Policy Optimization (TRPO)

(Schulman, Levine, et al. 2015)

Now consider parameterized policies π_θ , and uses a constraint on the KL divergence rather than a penalty¹.

In each iteration $k = 0, 1, 2 \dots$

$$\theta_{k+1} = \underset{\text{s.t. } D_{\text{KL}}(\pi_\theta(\cdot|s), \pi_{\theta_k}(\cdot|s)) \leq \delta}{\arg \max_\theta} \mathbb{E}_{s \sim \pi_{\theta_k}} [\sum_a \pi_\theta(s, a) A^{\pi_{\theta_k}}(s, a)]$$

In practice, use the sample-based estimation of $\mathbb{E}_{s \sim \pi_{\theta_k}}$ and $A^{\pi_{\theta_k}}$ and the sample-based estimation of average KL-divergence.

$$\theta_{k+1} = \underset{\text{s.t. } \hat{\mathbb{E}}_{s \sim \pi_{\theta_k}} [D_{\text{KL}}(\pi_\theta(\cdot|s), \pi_{\theta_k}(\cdot|s))] \leq \delta}{\arg \max_\theta} \hat{\mathbb{E}}_{s \sim \pi_{\theta_k}} \left[\sum_a \pi_\theta(s, a) \hat{A}^{\pi_{\theta_k}}(s, a) \right]$$

¹In practice, if using the penalty coefficient $\frac{2\gamma\varepsilon}{(1-\gamma)^2}$, the step sizes would be very small

Proximal Policy Optimization (PPO)

(Schulman, Wolski, et al. 2017)

Let $r_k(\theta) := \frac{\pi_\theta(s, a)}{\pi_{\theta_k}(s, a)}$, recall that TRPO maximize (under constraint)

$$\theta_{k+1} = \arg \max_{\theta} \hat{\mathbb{E}}_{s, a \sim \pi_{\theta_k}} \left[\frac{\pi_\theta(s, a)}{\pi_{\theta_k}(s, a)} \hat{A}^{\pi_{\theta_k}}(s, a) \right],$$

$$L^{CPI}(\theta) = \hat{\mathbb{E}}_k \left[r_k(\theta) \hat{A}_k \right]$$

But TRPO is hard to compute(), so an ad-hoc but efficient way comes:

PPO instead solve the unconstrained problem by maximize

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_k \left[\min \left(r_k(\theta) \hat{A}_k, \text{clip}(r_k(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_k \right) \right]$$

What does clip-then-min mean?

(Schulman, Wolski, et al. 2017)

$$L^{CLIP}(\theta) = \mathbb{E}_k \left[\min \left(r_k(\theta) \hat{A}_k, \text{clip}(r_k(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_k \right) \right]$$

- ▶ a pessimistic bound in the sense that
 - ▶ clip(ignore) when the policy improve
 - ▶ if $\hat{A} > 0$ and $r_k > 1 + \epsilon$, the final is $(1 + \epsilon)\hat{A}$
 - ▶ if $\hat{A} < 0$ and $r_k < 1 - \epsilon$, the final is $(1 - \epsilon)\hat{A}$
 - ▶ unclip when policy is worse
 - ▶ if $\hat{A} > 0$ and $r_k < 1 - \epsilon$, the final is $r_t \hat{A}$
 - ▶ if $\hat{A} < 0$ and $r_k > 1 + \epsilon$, the final is $r_k \hat{A}$

A soft way of conservative-greedy. By maximizing a pessimistic lower bound on the surrogate objective, PPO avoids the large jump to an estimated-better but actually-worse policy.

Proximal Policy Optimization (PPO)

(Schulman, Wolski, et al. 2017)

alternate between sampling data from the policy and performing several epochs of optimization on the sampled data

1. Initialize policy π^0
2. Repeat for $k=1,2,\dots$ until convergence
 - ▶ (policy evaluation) Play policy π from starting state distribution μ to generate sample trajectories. Estimate advantage function $\hat{A}^\pi(s, a)$ (by Monte Carlo or TD-learning)
 - ▶ (policy improvement) Optimize surrogate $L(\theta)$, with K epochs and minibatch size $M \leq NT$

$$\theta_{k+1} \leftarrow \theta$$

Outline

Intro

Smart-conservative-greedy parameterized algorithm

Intro

(2002) Conservative policy iteration

(2015) Trust Region Policy Optimization (TRPO)

(2017) Proximal Policy Optimization (PPO)

Policy learning for LLM

Intro

(2022) Reinforcement Learning with Human Feedback (RLHF)

(2023) Direct Policy Optimization (DPO)

(2024) Group Relative Policy Optimization (GRPO)

Beyond MDP

(2021) Decision Transformer

Discussion

On/Off-line and On/Off-policy

Intro - Policy learning for LLM

Reinforcement Learning with Human Feedback (RLHF)

(Christiano et al. 2017)(Ouyang et al. 2022)

- ▶ LLM is essentially a policy
 - ▶ if you think of states as context, action as next token
 - ▶ so policy learning methods like RL can be used for LLMs
 - ▶ but how? (reward?)
- ▶ Defining and Estimating reward
 - ▶ insight: humans are bad at scoring, but good at comparing
 - ▶ collect comparison data, and train a reward model (in (Ouyang et al. 2022), they use a 6B reward model)
 - ▶ r_{ϕ}^{BT} (prompt, output) is used in terminal reward when $t = T$
- ▶ Then apply RL! by PPO, etc ...

If you take a bandit environment view, it presents a random user prompt and expects a response to the prompt.

A special point of RL for LLM

How to get an RL-sense-better but not over-optimized model? i.e. how to make the final model not far way from the original π_{ref} ?

- ▶ per-token KL penalty into reward (sparse terminal reward version)

$$\log \frac{\pi_{\theta}(y \mid x)}{\pi_{\text{ref}}(y \mid x)} = \sum_{t=1}^T \log \frac{\pi_{\theta}(y_t \mid s_t)}{\pi_{\text{ref}}(y_t \mid s_t)}$$

$$R_t = 0, \quad t < T \quad R_T = r_{\phi}(x, y) - \beta \log \frac{\pi_{\theta}(y \mid x)}{\pi_{\text{ref}}(y \mid x)}$$

From finite-horizon RL to the KL-regularized RLHF objective

$$\max_{\pi_{\theta}} \mathbb{E}_{s_t \sim \rho, a_t \sim \pi_{\theta}(\cdot | s_t)} \left[\sum_{t=0}^T \gamma^t R(s_t, a_t) \right] \quad (\text{classic finite horizon RL goal})$$

$$= \max_{\pi_{\theta}} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta}(y|x)} [R(x, y)] \quad (\text{reduced to one-step contextual bandit})$$

$$= \max_{\pi_{\theta}} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta}(y|x)} \left[r_{\phi}(x, y) - \beta \log \frac{\pi_{\theta}(y|x)}{\pi_{\text{ref}}(y|x)} \right] \quad (\text{restriction into reward})$$

$$= \max_{\pi_{\theta}} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta}(y|x)} [r_{\phi}(x, y)] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(y \mid x) \parallel \pi_{\text{ref}}(y \mid x)]$$

Direct Policy Optimization

(Rafailov et al. 2023)

'Skip' the reward function

Original goal:

$$\mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta}(y|x)} [r_{\phi}(x, y)] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(y | x) \parallel \pi_{\text{ref}}(y | x)]$$

DPO goal:

$$\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_{\theta}(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right) \right]$$

Group Relative Policy Optimization (Shao et al. 2024)

Outline

Intro

Smart-conservative-greedy parameterized algorithm

Intro

(2002) Conservative policy iteration

(2015) Trust Region Policy Optimization (TRPO)

(2017) Proximal Policy Optimization (PPO)

Policy learning for LLM

Intro

(2022) Reinforcement Learning with Human Feedback (RLHF)

(2023) Direct Policy Optimization (DPO)

(2024) Group Relative Policy Optimization (GRPO)

Beyond MDP

(2021) Decision Transformer

Discussion

On/Off-line and On/Off-policy

Decision Transformer

a new framework for offline RL task

Decision Transformer(DT) reformulates the offline RL task as a **sequence modeling problem**.

- ▶ the return-to-go(sum of future rewards) $\hat{R}_t = \sum_{t'=t}^T r_{t'}$
 - ▶ reformulated trajectory: $(\hat{R}_1, s_1, a_1, \dots, \hat{R}_T, s_T, a_T)$
- ▶ Goal: learn a prediction model that predicts the action based on past returns-to-go, states, and actions(analogous to a language modeling task! action = next word)

$$\hat{a}_t = f_{\theta}(\hat{R}_{\leq t}, s_{\leq t}, a_{< t})$$

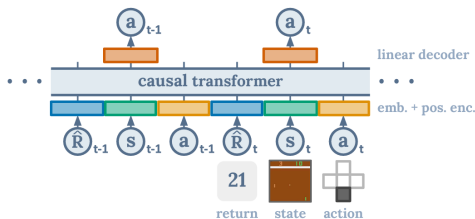


Figure: Decision Transformer architecture

Outline

Intro

Smart-conservative-greedy parameterized algorithm

Intro

(2002) Conservative policy iteration

(2015) Trust Region Policy Optimization (TRPO)

(2017) Proximal Policy Optimization (PPO)

Policy learning for LLM

Intro

(2022) Reinforcement Learning with Human Feedback (RLHF)

(2023) Direct Policy Optimization (DPO)

(2024) Group Relative Policy Optimization (GRPO)

Beyond MDP

(2021) Decision Transformer

Discussion

On/Off-line and On/Off-policy

Discussion

What does your policy look like?

Online v.s. Offline, On-policy v.s. Off-policy

- ▶ Online RL v.s. Offline RL: can/can not interact with the environment to get new sample trajectory [?]

references

- Christiano, Paul F et al. (2017). “Deep reinforcement learning from human preferences”. In: *Advances in neural information processing systems* 30.
- Kakade, Sham and John Langford (2002). “Approximately optimal approximate reinforcement learning”. In: *Proceedings of the nineteenth international conference on machine learning*, pp. 267–274.
- Ouyang, Long et al. (2022). “Training language models to follow instructions with human feedback”. In: *Advances in neural information processing systems* 35, pp. 27730–27744.
- Rafailov, Rafael et al. (2023). “Direct preference optimization: Your language model is secretly a reward model”. In: *Advances in neural information processing systems* 36, pp. 53728–53741.
- Schulman, John, Sergey Levine, et al. (2015). “Trust region policy optimization”. In: *International conference on machine learning*. PMLR, pp. 1889–1897.
- Schulman, John, Filip Wolski, et al. (2017). “Proximal policy optimization algorithms”. In: *arXiv preprint arXiv:1707.06347*.
- Shao, Zhihong et al. (2024). “Deepseekmath: Pushing the limits of mathematical reasoning in open language models”. In: *arXiv preprint arXiv:2402.03300*.